Vishal Mathad

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Education

Army Institute of Technology **Aug. 2023 – June 2027**
Bachelor of Engineering in Electronics and Telecommunication Engineering, CGPA: 8.23/10 *Pune, Maharashtra*

Army Public School **2021 – 2022**
Higher Secondary Education, Percentage: 90% *Pathankot, Punjab*

Technical Skills

Languages: C/C++, Python, JavaScript, Go (Golang), HCL, HTML/CSS
Backend & Web: Node.js, Express.js, FastAPI, net/http, Kafka
Databases: SQL, PostgreSQL, MongoDB
DevOps & Cloud: Linux, Docker, Kubernetes, Jenkins, CI/CD, Terraform, AWS, Prometheus, Grafana
CS Fundamentals: Data Structures & Algorithms, Database Management System, Operating Systems, Computer Networks, Object Oriented Programming, System Design
Developer Tools: Git, GitHub, Postman, VS Code, Figma

Projects

Chaos-Driven Self-Healing Kubernetes Platform **Sept 2026**
Go, Node.js, Docker, Kubernetes/KIND, Helm, Prometheus/Grafana, GitHub Actions, AWS

- Built a chaos engineering platform with a Go controller and Node.js/Express application to inject five failure scenarios: pod eviction, application health failure, readiness failure, latency injection, and CPU stress; measured recovery duration and validated application resilience.
- Deployed on a 3-node KIND cluster using Helm, configuring startup/liveness/readiness probes, rolling updates, resource limits, a PodDisruptionBudget (minimum 2 available replicas), and HPA-based autoscaling from 3–6 replicas at 60% CPU utilization.
- Automated CI/CD with GitHub Actions, integrating Node.js and Go tests, static analysis, Trivy vulnerability scanning, and commit-SHA-tagged GHCR images; enabled AWS OIDC authentication and SSM-based EC2 deployments with rollout and health verification.

Git Hosting Server over SSH **Oct 2026**
Go, PostgreSQL, SSH, Git

- Built a local Git hosting server in Go over SSH, supporting native clone, fetch, pull, and push with PostgreSQL-backed users, SSH keys, repositories, and READ/WRITE permissions.
- Implemented strict Git command allowlisting, repository-name validation, symlink-aware path confinement, direct Git subprocess execution, and connection-scoped cancellation without shell-based command execution.
- Added structured logging and CLI-based administration with security, fuzz, concurrency, cancellation, and end-to-end acceptance tests for authentication, authorization, repository isolation, and Git operations.

Leadership Experience

Spiritual Club **April 2025 – April 2026**
Secretary *Army Institute of Technology*

- Led the planning and execution of 10+ student events across the academic year, coordinating teams, logistics, and event operations.
- Collaborated with Heartfulness to organize a meditation camp for 200+ students, contributing to a 30% increase in participation in wellness activities.

Achievements & Extra-Curricular

- Participated in multiple chess tournaments and achieved a peak rating of 1592.
- Solved 400+ programming problems across LeetCode, Codeforces, and GeeksforGeeks.
- Awarded the PMSS Scholarship for academic excellence.
- Attended the 3-day International Youth Summit, "Rising With Kindness," at Kanha Shanti Vanam organized by Heartfulness.
- Participated in seminars focused on financial literacy and personal finance.